

FMCG Scope 3 Optimization & Strategy Framework

A GHG-Compliant Data Modeling
& Decarbonization Framework
(Hup Seng Case Study)

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Introduction

About This Portfolio Case Study

This document is an independent portfolio case study designed to demonstrate my capabilities in ESG data modeling, Scope 3 accounting, and strategic sustainability planning. Hup Seng Industries Berhad was selected as a representative case of Malaysia's local FMCG sector to explore practical solutions for common data transparency challenges. This project is strictly non-commercial, created solely for educational and portfolio presentation purposes, and does not represent an official report from or affiliation with Hup Seng Industries Berhad. Project Completed: July 2026

How to Read This Document

To help you navigate this case study efficiently, the core insights are organized into three sequential parts:

1. The Data Gap (Pg 4-5): An analysis of the baseline Scope 3 emissions data, highlighting the initial "data black box" challenge and how standard-compliant data modeling is applied to fill critical missing gaps.
2. Data Strategy & Architecture (Pg 6-7): A long-term three-tier data capture framework built to transition secondary estimates into audit-ready primary supplier data.
3. Strategic Roadmap (Pg 8): A practical, life-cycle decarbonization roadmap designed across four operational pillars, connecting logical sustainability actions to real-world business contexts.



Baseline Analysis – Business Footprint & Value Chain

1. Data Source & Objective

The operational baseline below is extracted directly from official disclosures—the Hup Seng’s Sustainability Report FY2025 and Annual Report 2025. Mapping Hup Seng’s physical manufacturing footprint, sales network, and supplier distribution provides the necessary foundation to identify where emissions naturally concentrate across its value chain.

2. Operational Footprint Overview

Supplier Base

Approximately 97.1% of total procurement spend is concentrated with local Malaysian suppliers.

	FY2025
Proportion of spending on local suppliers (%)	97.1

Sales Network

Strong domestic penetration via direct distribution networks in Peninsular Malaysia and wholesale partners across East Malaysia and export markets.

Region_Market	Revenue_Share(%)	Sales_Channel_Detail
Domestic (East and West Malaysia)	81.1	For West Malaysia: Direct sales through 6 branches (Klang Valley, Kota Bharu, Kuantan, Ipoh, Butterworth, Alor Setar); Hypermarkets, Retailers, Wholesalers, Convenience Stores. For East Malaysia: Appointed Distributors
Export (approximately 40 countries)	18.9	Asia market (83.3% of exports) including Indonesia, Singapore, Thailand, Myanmar, Saudi Arabia

Production Bases (factory)

In-house manufacturing centered in Johor, Malaysia providing strong operational control.

Operating_Subsiary	Product_Line	Factory_Location
Hup Seng Perusahaan Makanan (M) Sdn Bhd	Biscuits	Tongkang Pecah, Batu Pahat, Johor
In-Comix Food Industries Sdn Bhd	Beverages	Senai, Johor

Core Product Lines

Biscuit manufacturing drives group revenues (94.8% of total turnover).

Category	Product_Line	Revenue_Share(%)
Biscuits	Hup Seng Cream Crackers	63.2
Biscuits	Marie Biscuits, Sugar Crackers, Chocolate Teddy Biscuits, Chibi Biscuits	31.6
Beverages and other Agent Products	In-Comix (3-in-1 Instant Coffee, Instant Tea, Nutritious Cereal)	5.2

3. Business Chain Synthesis

Hup Seng operates a localized, vertically integrated business model. It combines strong in-house manufacturing with a high-control domestic distribution network. From upstream raw material procurement (flour, palm oil, packaging) to midstream baking operations and downstream logistics, the vast majority of its business chain resides locally within Malaysia.

The Problem – Scope 3 Data Black Box

1. Core Categories Mapping (GHG Protocol Alignment)

By cross-referencing Hup Seng’s business footprint with the GHG Protocol Scope 3 Standard, 8 out of 15 categories were identified as core areas requiring formal disclosure:

- Upstream: Cat 1 (Purchased Goods), Cat 4 (Upstream Transport), Cat 5 (Operational Waste)
- Operations & Commute: Cat 6 (Business Travel), Cat 7 (Employee Commuting)
- Downstream: Cat 9 (Downstream Transport), Cat 11 (Use of Sold Products), Cat 12 (End-of-Life Treatment)

2. Core Analysis & Insights

Observation:

- Out of 8 core categories, Hup Seng only disclosed Category 6 & 7 (Business Travel & Employee Commuting) at **2,083 tCO2e**. The remaining 6 core categories contain zero data (NULL).

Information:

- The current baseline reveals a massive **Scope 3 data black box** across the group's core operations.

Insight:

- Given the company's model of concentrated manufacturing and local distribution, it is reasonable to infer that the bulk of actual emissions resides upstream in Category 1 (Raw Materials) and downstream in Categories 4 & 9 (Logistics). The disclosed Categories 6 & 7 represent an insignificant fraction of total operations, leaving the true environmental footprint unmeasured.

3. Commercial & Compliance Risks

If left unresolved, this Scope 3 data gap creates immediate strategic risks:

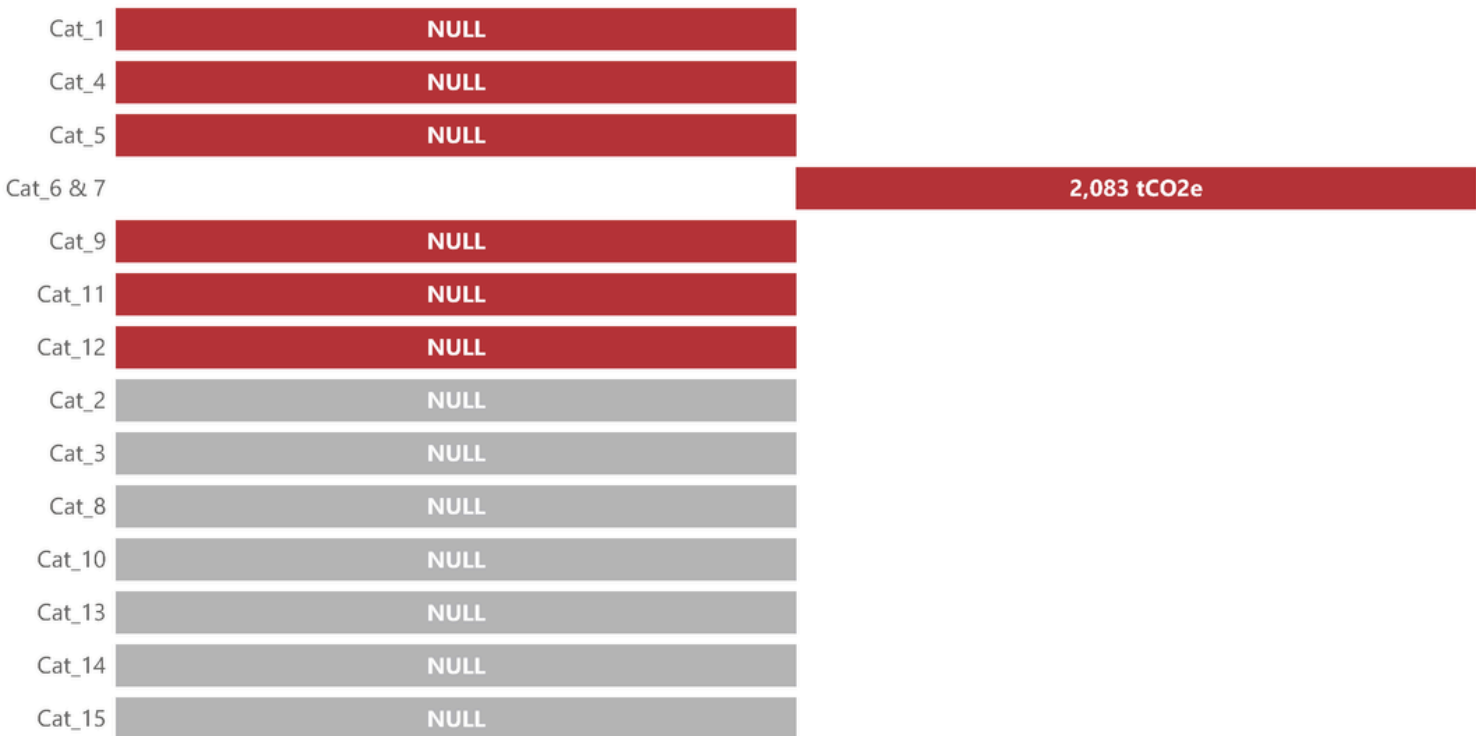
- **Strategic Blind Spots:** Inability to pinpoint emission hotspots, making it impossible to design an actionable or cost-effective Decarbonization Roadmap.
- **Compliance & Rating Penalties:** Slow progress in Scope 3 readiness risks non-compliance with Bursa Malaysia’s CSI platform standards, leading to stagnant or downgraded ESG sustainability ratings.
- **Financial Impairment:** Lower sustainability performance directly exposes the business to investor caution, higher debt financing costs, and potential green tax liabilities.

4. Summary Conclusion

Hup Seng faces a critical Scope 3 data gap that obscures its real supply chain carbon footprint. Closing this black box is no longer just a reporting task—it is a strategic urgency to protect business valuation and market competitiveness.

Hup Seng Scope 3 Emissions 2025

Business_Scope ● A very small part or irrelevant ● Core



Note: Red Bars = Core Required Categories for formal disclosure

Short-Term Solution – Data Modeling & Gap Closure

1. Strategic Objective

Before long-term supplier engagement can take effect, companies need an immediate, standard-compliant way to quantify their emission baselines. The goal here is not to create ideal figures, but to replace complete blind spots with audit-ready estimates using recognized international protocols.

2. Methodological Approach & Technical Scope

To resolve core blind spots without compromising accounting integrity, targeted secondary estimation methodologies were applied strictly to primary manufacturing inputs and operational outputs:

- **Category 1 (Purchased Goods & Services):**

Quantified at **29,725 tCO₂e**. Rather than relying on high-level spend proxies, this baseline applies a Raw Material Mass Disaggregation Model—extracting physical production volumes of core biscuit ingredients (flour, palm oil, sugar) from financial disclosures and multiplying them by GHG Protocol-compliant secondary emission factors. (Note: Packaging materials are held for future primary data capture).

- **Category 5 (Waste Generated in Operations):**

Quantified at **15 tCO₂e** using production-waste intensity metrics aligned with the GHG Protocol Scope 3 Waste Treatment Selection Framework.

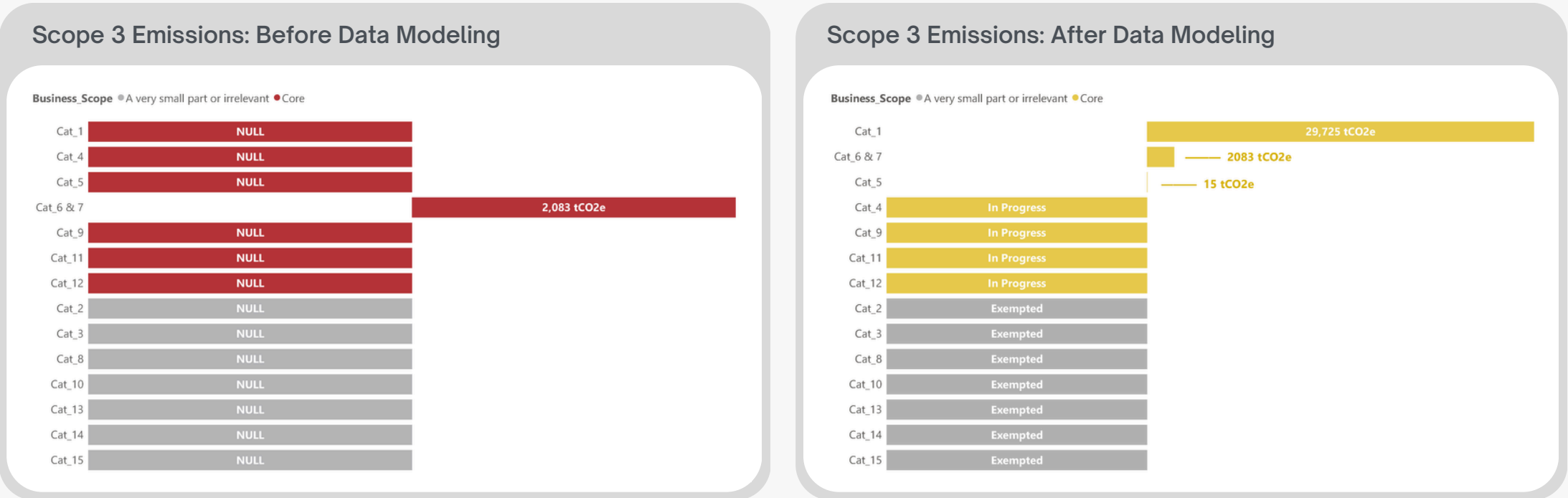
3. Professional Boundary & Transparency

Rather than over-extrapolating unverified figures, clear boundaries were established for the remaining categories:

- **In Progress (Cat 4, 9, 11, 12):** Temporarily held as active focus areas pending structured hybrid activity data from logistics partners and post-consumer waste stream models.
- **Exempted (Cat 2, 3, 8, 10, 13, 14, 15):** Formally evaluated and exempted due to non-materiality or non-applicability to Hup Seng’s core biscuit manufacturing business model.

4. Analytical Value

By applying standard-compliant data modeling, the immediate Scope 3 baseline expanded from a mere **2,083 tCO₂e** (business travel & employee commuting only) to a far more representative **31,823 tCO₂e**. This provides Hup Seng with its first actionable carbon baseline—establishing a realistic starting point for supply chain decarbonization.



**Full derivations, mathematical formulas, and standard-compliance details for all modeling methodologies are available via the link in the "Methodology & Deep-Dive" section of this portfolio.*

Mid-Term Strategy – Three-Tier Data Capture Strategy

1. Strategic Transition

While standard-compliant modeling (Page 6) closes some of the current data gaps, secondary estimates cannot support long-term decarbonization audits or supplier accountability. To achieve audit-grade transparency, Hup Seng must transition from secondary modeling to structured primary data capture across its supply chain.

2. Operational Mechanisms by Tier

To balance data precision with supplier execution feasibility, a differentiated three-tier capture architecture is established:



The Core Tier (Primary Data Source) — Categories 1 & 5

Mechanism:
Mandates contractual Primary Data collection clauses across core operations. For Category 1, requires the top 20% core raw material suppliers (flour, palm oil) to report line-level emissions; for long-tail suppliers, use the average emission intensity (tCO2e/million MYR output) collected from the core suppliers to simulate full coverage. For Category 5, updates vendor agreements to collect itemized primary waste logs directly from third-party recovery contractors.



The Bridge Tier (Hybrid / Activity Data Source) — Categories 4 & 9

Mechanism:
Integrates operational activity metrics into third-party logistics contracts. Carriers are required to append verified tonne-kilometer (t . km) metrics and vehicle profile details directly onto billing invoices.



The Long-Tail Tier (Extrapolated Secondary Data) — Categories 11 & 12

Mechanism:
Applies an Energy Intensity Extrapolation Model for consumer product usage (Cat 11) and a Geographically-Weighted Recycling Model for post-consumer tin/packaging end-of-life streams (Cat 12).

Overview Table of Data Capture & Gap-filling Methods

Capture_Point	Related_Categories	Industry_Chain_Position	Methods
Core materials	Cat 1	Upstream	Supplier-specific
Long-tail small supplier	Cat 1	Upstream	Weighted Emission Intensity Proxy
Production waste	Cat 5	Upstream	Supplier-specific
Export/Outbound Logistics	Cat 4 & 9	Upstream & Downstream	Activity-based model (distance x weight)
Use of sold products	Cat 11	Downstream	Use Profiles Model (Energy Intensity Extrapolation Method)
Product packaging endpoint	Cat 12	Downstream	Average-data Method Model (Local Recycle Rate Statistics)

**Detailed strategy operational mechanism, full derivations, mathematical formulas, and standard-compliance details are available via the link in the "Methodology & Deep-Dive" section of this portfolio.*

3. Governance & System Integration

Data capture is only as effective as the systems supporting it. To operationalize this framework, two key enablers are embedded into business workflows:

- Procurement Governance: Inclusion of ESG metrics in quarterly supplier scorecard reviews, aligning carbon transparency directly with procurement allocation decisions.
- CSI Platform Readiness: Structuring all collected data formats to align directly with Bursa Malaysia’s Centralised Sustainability Intelligence (CSI) platform, ensuring seamless regulatory compliance.

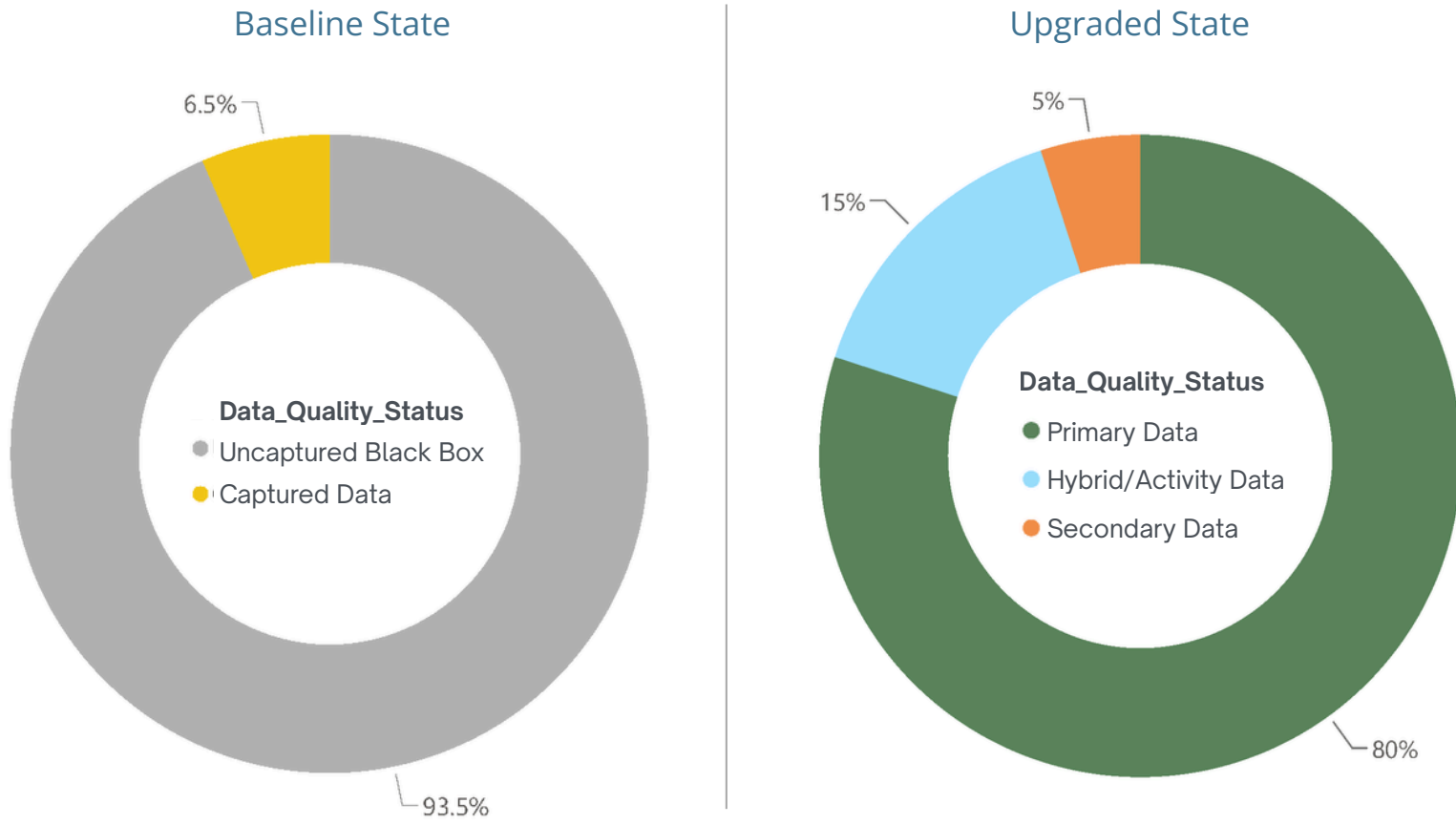
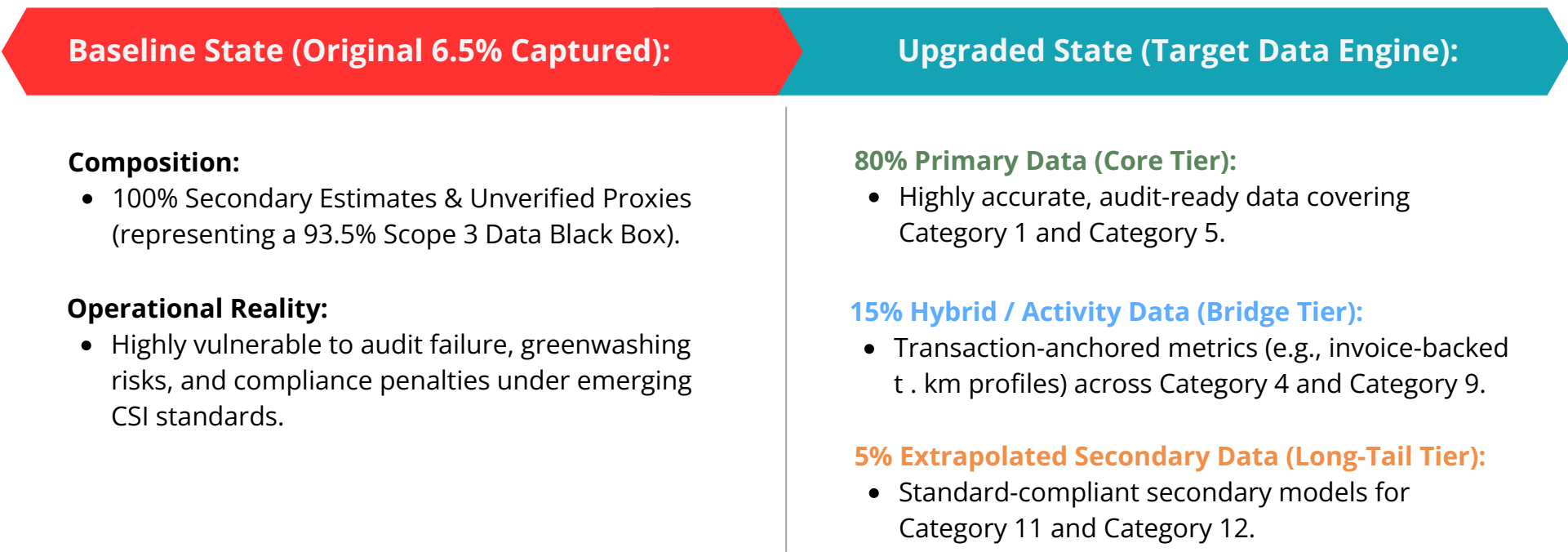
Data Quality Upgrade: Before vs. After

1. Strategic Context

Deploying the Three-Tier Framework (Page 7) fundamentally transforms Hup Seng’s data maturity. By moving away from random data collection, the organization establishes a risk-proportionate data engine that satisfies stringent regulatory audits without overwhelming administrative capacity.

2. Quantifiable Data Maturity Progression

Comparing the baseline state against the post-implementation architecture reveals a stark shift in data fidelity:



**Detailed derivation and standard compliance are available via the link in the "Methodology & Deep-Dive" section of this portfolio.*

3. Strategic Impact & ROI

This data transformation yields two critical business advantages:

- **Audit-Ready Compliance:** Upgrading 95% of the carbon inventory into Primary and Hybrid streams positions Hup Seng for frictionless assurance under Bursa Malaysia’s CSI framework and international rating criteria.
- **Resource Optimization:** Demonstrates that full supply chain transparency does not require chasing 100% of suppliers. By focusing procurement leverage on core nodes, the enterprise captures 80% of true carbon impact using just 20% of administrative bandwidth.

4. Summary Conclusion

By designing a smart, Pareto-aligned data capture architecture, Hup Seng transforms a compliance liability into a strategic advantage—proving that corporate climate readiness is a matter of intelligent system design, not brute-force data collection.

Decarbonization Roadmap Design

1. Strategic Context

With an audit-ready data baseline established (Page 8), emissions reduction shifts from theoretical targets to targeted lifecycle execution. Rather than applying generic carbon offsets, this strategic roadmap addresses carbon hot-spots across the four stages of Hup Seng's product lifecycle—balancing operational feasibility with long-term commercial value. *This page focuses on 'logic and lifecycle framework design', rather than the secondary modelling of simulated data.*

2. Tactical Execution Across the 4 Lifecycle Pillars

**Detailed strategic mechanisms are available via the link in the "Methodology & Deep-Dive" section of this portfolio.*



3. Strategic Value & Implementation Philosophy

This roadmap deliberately avoids speculative carbon credit purchases. By embedding decarbonization directly into raw material purchasing, factory waste loops, iconic packaging returns, and consumer habits, Hup Seng strengthens core supply chain resilience while future-proofing its brand equity against regulatory and market shifts.

Model Scalability & Alignment

1. Macro Strategic Value: From Single Case to Industry Blueprint

The framework developed in this study goes beyond a single-company solution for Hup Seng. By integrating **Secondary Data Modeling (Page 6)**, a **Three-Tier Capture Strategy (Pages 7 & 8)**, and a **Lifecycle Roadmap (Page 9)**, this project establishes a pragmatic, highly scalable blueprint. It offers a practical entry point for mid-sized FMCG and manufacturing enterprises across Malaysia seeking to solve Scope 3 data paralysis without excessive operational burden.

2. National Policy & Regulatory Alignment

To ensure long-term commercial viability and regulatory compliance, the overall methodology is mapped directly against Malaysia’s primary sustainability and economic policy frameworks:

**BURSA MALAYSIA
CSI PLATFORM
ALIGNMENT**

Integration:
Direct mapping of Tier 1 & Tier 2 emission metrics into standard disclosure formats, enabling frictionless data integration into Bursa's Centralised Sustainability Intelligence (CSI) platform.

**CMM SIMPLIFIED
ESG DISCLOSURE
GUIDE (SEDG)
FOR SMES**

Integration:
Adopts the "Basic to Medium" progression tiers recommended by Capital Markets Malaysia (CMM), allowing supply chain partners to build ESG readiness at an achievable pace.

**NEW INDUSTRIAL
MASTER PLAN
(NIMP) 2030**

Integration:
Supports **Mission 3 (Push for Net Zero)** by transforming traditional food manufacturing supply chains into digitally trackable, low-carbon industrial ecosystems.

3. Replicability & Strategic ROI

The core strength of this system design lies in its commercial realism. It proves that corporate climate action does not require immediate, capital-intensive infrastructure overhauls or exhaustive supplier surveys.

- High Deployment Speed**
 - Pre-structured estimation models allow companies to close critical data black boxes in weeks rather than years.
- Cost-Effective Scalability**
 - Focusing administrative bandwidth strictly on high-impact core nodes delivers maximum compliance coverage with minimal overhead.
- Capital Protection**
 - Preempts emerging green tax liabilities, lowers loan pricing risks, and safeguards investor confidence across regional markets.

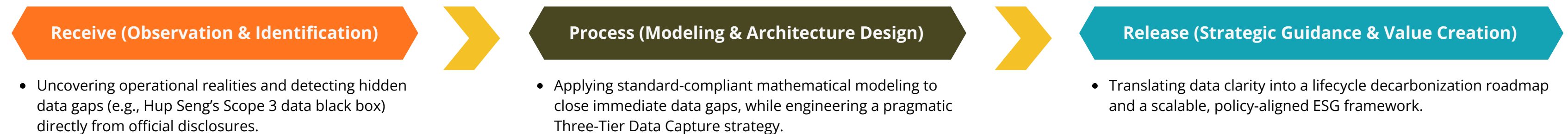
4. Executive Summary Conclusion

By combining rigorous accounting protocols with practical business workflows, this framework transforms Scope 3 reporting from a complex compliance hurdle into a clear strategic advantage—proving that sustainable industrial transformation in Malaysia is both achievable and commercially sound.

Personal Methodology & Deep-dive Appendix

1. Personal Cognitive Framework

This project reflects my core analytical compass—translating complex, fragmented inputs into structured, actionable enterprise logic through three distinct phases:



2. Technical Toolkit & Domain Standards

- **ESG & Standard Frameworks:** GHG Protocol Corporate Value Chain Standard, Lifecycle Assessment (LCA) Framework, CMM SEDG.
- **Data Extraction & Engineering:** Gemini Notebook for automated document extraction, Excel for data management & modeling.
- **Data Analytics & Visualization:** Power BI for data visualization, Canva and Notion for portfolio production.

3. Deep-Dive Appendix & Technical Logs

To explore the complete derivation models, mathematical formulations, and underlying code, access the full interactive Notion Appendix below:

[!\[\]\(17413706fd4997a1a4bdf85c6864eee1_img.jpg\) FMCG-Scope-3-Strategy-Framework-Technical-Appendix](#)

What's inside the full appendix:

- Methodology Derivations: Step-by-step mathematical proofs for Category 1, 5, 11 & 12 estimation models.
- Technical Engine Logs: Data capture workflows and data transformation scripts.

Closing Note

Data analytics is most powerful when it drives practical, real-world solutions. Thank you for reading through this study.